

CUSTOMER SEGMENTATION

Synent Technologies Data Science Internship

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Objective

The objective of this project is to group customers into meaningful segments based on their annual income and spending behavior using K-Means Clustering.

Dataset Information

Dataset Name: Mall Customer Dataset

Source: Kaggle

Rows: 200

Columns: 5

Features:

- CustomerID
- Gender
- Age
- Annual Income (k\$)
- Spending Score (1-100)
- Methodology

The following steps were performed:

- Loaded the dataset using Pandas
- Examined dataset structure and missing values
- Selected Annual Income and Spending Score for clustering
- Applied the Elbow Method to determine the optimal number of clusters
- Applied K-Means Clustering
- Visualized customer segments using scatter plots
- Generated customer behavior insights
- Results

The analysis identified five customer segments with different spending and income characteristics.

Key Findings:

- High-income, high-spending customers represent premium customers.

- High-income, low-spending customers provide opportunities for targeted marketing.
- Moderate-income customers form a stable customer group.
- Low-income, low-spending customers require different engagement strategies.
- Customer segmentation supports personalized marketing and business decision-making.

Generated Visualizations:

- Elbow Method Graph
- Customer Cluster Visualization
- Cluster Insights Chart
- Conclusion

Customer Segmentation using K-Means Clustering effectively grouped customers according to their behavior patterns. The resulting segments can help businesses optimize marketing efforts, improve customer retention, and increase overall profitability.